

Energy in the electric field: work done to assemble charges

Work to bring a test charge in a given electric field

Suppose an external agent slowly brings a (positive) test charge q from point $\mathbf{r}_0$ to $\mathbf{r}_1$ along any path C in an electrostatic field $\mathbf{E}(\mathbf{r})$. The infinitesimal work done by the external agent is

$$dW_{\text{ext}} = \mathbf{F}_{\text{ext}} \cdot d\mathbf{l} = (-\mathbf{F}_{\text{elec}}) \cdot d\mathbf{l} = (-q\mathbf{E}) \cdot d\mathbf{l}.$$

Hence,

$$W_{\text{ext}}(\mathbf{r}_0 \rightarrow \mathbf{r}_1) = -q \int_C \mathbf{E} \cdot d\mathbf{l}. \quad (1)$$

Using the definition of potential difference,

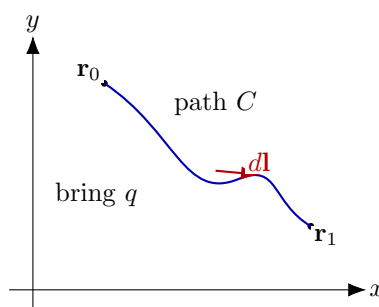
$$\phi(\mathbf{r}_1) - \phi(\mathbf{r}_0) = - \int_C \mathbf{E} \cdot d\mathbf{l},$$

equation (1) becomes

$$W_{\text{ext}}(\mathbf{r}_0 \rightarrow \mathbf{r}_1) = q[\phi(\mathbf{r}_1) - \phi(\mathbf{r}_0)]. \quad (2)$$

If we choose the reference $\phi(\infty) = 0$ and bring the charge from infinity,

$$W_{\text{ext}}(\infty \rightarrow \mathbf{r}) = q\phi(\mathbf{r}). \quad (3)$$



Energy required to bring N charges from infinity (assembly work)

Now we build a configuration of N point charges $Q_1, Q_2, \dots, Q_N$ by bringing them one-by-one from infinity to fixed locations

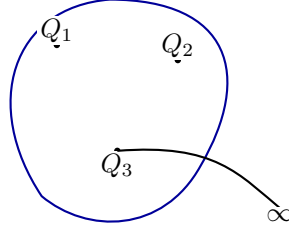
$$\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N.$$

We choose $\phi(\infty) = 0$.



Step 1: Place Q_1 at $\mathbf{r}_1$. With no charges present initially, the potential everywhere is zero. Therefore,

$$W_1 = Q_1 \phi_0(\mathbf{r}_1) = 0.$$



Step 2: Place Q_2 at $\mathbf{r}_2$. When only Q_1 is already present, the potential at $\mathbf{r}_2$ due to Q_1 is

$$\phi_1(\mathbf{r}_2) = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{|\mathbf{r}_2 - \mathbf{r}_1|}.$$

Hence the work to bring Q_2 from infinity is

$$W_2 = Q_2 \phi_1(\mathbf{r}_2) = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{|\mathbf{r}_2 - \mathbf{r}_1|}.$$

Step 3: Place Q_3 at $\mathbf{r}_3$. Now the potential at $\mathbf{r}_3$ due to the already-placed charges Q_1, Q_2 is

$$\phi_2(\mathbf{r}_3) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{|\mathbf{r}_3 - \mathbf{r}_1|} + \frac{Q_2}{|\mathbf{r}_3 - \mathbf{r}_2|} \right),$$

so

$$W_3 = Q_3 \phi_2(\mathbf{r}_3) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1 Q_3}{|\mathbf{r}_3 - \mathbf{r}_1|} + \frac{Q_2 Q_3}{|\mathbf{r}_3 - \mathbf{r}_2|} \right).$$

General step: place Q_i at $\mathbf{r}_i$. When $Q_1, \dots, Q_{i-1}$ are already present,

$$\phi_{i-1}(\mathbf{r}_i) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^{i-1} \frac{Q_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad W_i = Q_i \phi_{i-1}(\mathbf{r}_i).$$

Therefore, the *total work* to assemble all N charges is

$$\boxed{W = \sum_{i=1}^N W_i = \sum_{i=1}^N Q_i \left[\frac{1}{4\pi\epsilon_0} \sum_{j=1}^{i-1} \frac{Q_j}{|\mathbf{r}_i - \mathbf{r}_j|} \right]}. \quad (4)$$

Key idea (what the double sum means)

- Each W_i is the work to bring in the i th charge against the potential created by the *previously placed* charges.
- The inner sum $\sum_{j=1}^{i-1}$ counts only interactions with charges already present, so no term is double-counted.

Symmetric (pairwise) form of the energy

Expression (4) counts each interacting pair exactly once (with $j < i$). Hence we can rewrite it as a clean pair-sum:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{1 \leq j < i \leq N} \frac{Q_i Q_j}{|\mathbf{r}_i - \mathbf{r}_j|} = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{\substack{i,j=1 \\ i \neq j}}^N \frac{Q_i Q_j}{|\mathbf{r}_i - \mathbf{r}_j|}. \quad (5)$$

The factor $1/2$ in the last form appears because the sum over $i \neq j$ counts each pair twice: (i, j) and (j, i) .

Final result

For N point charges assembled quasistatically from infinity,

$$W = \frac{1}{4\pi\epsilon_0} \sum_{1 \leq j < i \leq N} \frac{Q_i Q_j}{|\mathbf{r}_i - \mathbf{r}_j|}$$

is the electrostatic energy stored in the configuration (equal to the work done by the external agent).

Energy required to build a continuous charge distribution $\rho(\mathbf{r})$

Motivation (why we need a new idea)

For discrete charges, we built the configuration by bringing charges one-by-one from infinity and summed the work. For a *continuous* distribution $\rho(\mathbf{r})$, we cannot add “one charge” at a time. Instead, we imagine *building the distribution gradually* by scaling up the charge density while keeping the *same shape* of $\rho(\mathbf{r})$.

Gradual build-up (the scaling idea in the sketch)

Let the final (target) charge distribution be $\rho(\mathbf{r})$ supported inside some region V (same shape throughout the process).



We introduce a build parameter $\lambda \in [0, 1]$ and define an intermediate distribution

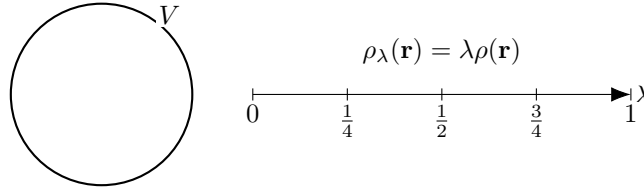
$$\boxed{\rho_\lambda(\mathbf{r}) = \lambda \rho(\mathbf{r}), \quad 0 \leq \lambda \leq 1.} \quad (6)$$

So,

$$\lambda = 0 : \rho_0(\mathbf{r}) = 0 \quad (\text{no charge}), \quad \lambda = 1 : \rho_1(\mathbf{r}) = \rho(\mathbf{r}) \quad (\text{final}).$$

This is exactly what your discrete-time sketch is conveying: at step k out of N steps,

$$\rho_k(\mathbf{r}) = \frac{k}{N} \rho(\mathbf{r}).$$



Incremental work at an intermediate stage

At a given stage λ , the charges already present create an electrostatic potential $\phi_\lambda(\mathbf{r})$.

Now we increase λ by a small amount $d\lambda$. The small added charge element is

$$d\rho = \frac{\partial \rho_\lambda}{\partial \lambda} d\lambda = \rho(\mathbf{r}) d\lambda, \quad dq = d\rho dV = \rho(\mathbf{r}) d\lambda dV.$$

To bring this added charge in from infinity and place it at position $\mathbf{r}$, the required incremental work is

$$dW = \int_V \phi_\lambda(\mathbf{r}) dq = \int_V \phi_\lambda(\mathbf{r}) \rho(\mathbf{r}) d\lambda dV.$$

Hence,

$$\boxed{\frac{dW}{d\lambda} = \int_V \phi_\lambda(\mathbf{r}) \rho(\mathbf{r}) dV.} \quad (7)$$

Key observation: potential scales linearly with the amount of charge

Electrostatics is linear: if we scale the charge density by λ , the resulting potential also scales by λ :

$$\boxed{\phi_\lambda(\mathbf{r}) = \lambda \phi(\mathbf{r}),} \quad (8)$$

where $\phi(\mathbf{r})$ is the potential due to the final distribution $\rho(\mathbf{r})$.

Substitute $\phi_\lambda = \lambda \phi$ into (7):

$$\frac{dW}{d\lambda} = \int_V (\lambda \phi(\mathbf{r})) \rho(\mathbf{r}) dV = \lambda \int_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV.$$



Integrate from $\lambda = 0$ to 1:

$$W = \int_0^1 \frac{dW}{d\lambda} d\lambda = \left(\int_0^1 \lambda d\lambda \right) \left(\int_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV \right) = \frac{1}{2} \int_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV.$$

Final result (continuous-charge energy)

$$W = \frac{1}{2} \int_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV. \quad (9)$$

The factor $\frac{1}{2}$ appears because while building the charge up, the potential is not at its final value initially; it ramps linearly from 0 to ϕ .

Equivalent double-integral form (purely in terms of ρ)

Using

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV',$$

equation (9) becomes

$$W = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \int_V \int_V \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV dV'. \quad (10)$$

Energy to build a continuous charge distribution: discrete build-up (k/N) viewpoint

Let the final (target) charge distribution be $\rho(\mathbf{r})$ supported in a region V . We imagine “building” this distribution in N equal steps while keeping the *shape* fixed.

Step 1: Charge and potential at the k -th stage

At the k -th stage ($k = 0, 1, 2, \dots, N$), the charge density is

$$\rho_k(\mathbf{r}) = \frac{k}{N} \rho(\mathbf{r}). \quad (11)$$

The electrostatic potential produced by ρ_k is

$$\begin{aligned}
 \phi_k(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho_k(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV' \\
 &= \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\frac{k}{N}\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV' \\
 &= \frac{k}{N} \underbrace{\left(\frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV' \right)}_{\phi(\mathbf{r})} \\
 &= \boxed{\frac{k}{N} \phi(\mathbf{r})},
 \end{aligned} \tag{12}$$

where $\phi(\mathbf{r})$ is the potential due to the *final* distribution $\rho(\mathbf{r})$.

Step 2: Work for adding the next “slice” of charge

From stage k to stage $k + 1$, we add

$$\Delta\rho(\mathbf{r}) = \rho_{k+1}(\mathbf{r}) - \rho_k(\mathbf{r}) = \frac{1}{N}\rho(\mathbf{r}), \quad dq = \Delta\rho(\mathbf{r}) dV = \frac{\rho(\mathbf{r})}{N} dV. \tag{13}$$

At this moment, the charge already present creates potential $\phi_k(\mathbf{r}) = \frac{k}{N}\phi(\mathbf{r})$. Hence the incremental work is (“charge $\times$ potential”)

$$\begin{aligned}
 \Delta W_k &= \iiint_V \phi_k(\mathbf{r}) dq \\
 &= \iiint_V \left(\frac{k}{N}\phi(\mathbf{r}) \right) \left(\frac{\rho(\mathbf{r})}{N} dV \right) \\
 &= \boxed{\frac{k}{N^2} \iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV}.
 \end{aligned} \tag{14}$$

Step 3: Sum over all steps

Total work:

$$\begin{aligned}
 W &= \sum_{k=0}^{N-1} \Delta W_k = \left(\iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV \right) \frac{1}{N^2} \sum_{k=0}^{N-1} k \\
 &= \left(\iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV \right) \frac{1}{N^2} \cdot \frac{(N-1)N}{2} \\
 &= \frac{N-1}{2N} \iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV.
 \end{aligned} \tag{15}$$

Taking $N \rightarrow \infty$ (infinitely fine build-up),

$$W = \frac{1}{2} \iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV. \quad (16)$$

Why the $\frac{1}{2}$ appears (one-line interpretation)

During the build-up, the potential increases linearly from 0 to $\phi(\mathbf{r})$, so the *average* potential seen by added charge is $\phi/2$, giving the factor $\frac{1}{2}$.

Intuitive origin of the $\frac{1}{2}$ factor (triangle picture)

Let the final (target) charge distribution be $\rho(\mathbf{r})$ with potential $\phi(\mathbf{r})$. We build it up gradually by a scale factor $\lambda \in [0, 1]$:

$$\rho_\lambda(\mathbf{r}) = \lambda \rho(\mathbf{r}) \implies \phi_\lambda(\mathbf{r}) = \lambda \phi(\mathbf{r}),$$

because electrostatics is linear in charge.

Key idea. When we add a small additional fraction $d\lambda$ of charge, the potential that this newly added charge “feels” is only the *already-built* potential $\phi_\lambda = \lambda\phi$ (not the full ϕ). Hence the incremental work is

$$dW = \iiint_V \phi_\lambda(\mathbf{r}) d\rho(\mathbf{r}) dV = \iiint_V (\lambda\phi(\mathbf{r})) (\rho(\mathbf{r}) d\lambda) dV = \lambda \left(\iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV \right) d\lambda.$$

Define the constant

$$I \triangleq \iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV.$$

Then

$$\frac{dW}{d\lambda} = \lambda I,$$

which is a straight line in λ . Therefore the total energy is the *area under this line* from $\lambda = 0$ to 1:

$$W = \int_0^1 \lambda I d\lambda = \frac{1}{2} I = \boxed{\frac{1}{2} \iiint_V \rho(\mathbf{r}) \phi(\mathbf{r}) dV}.$$

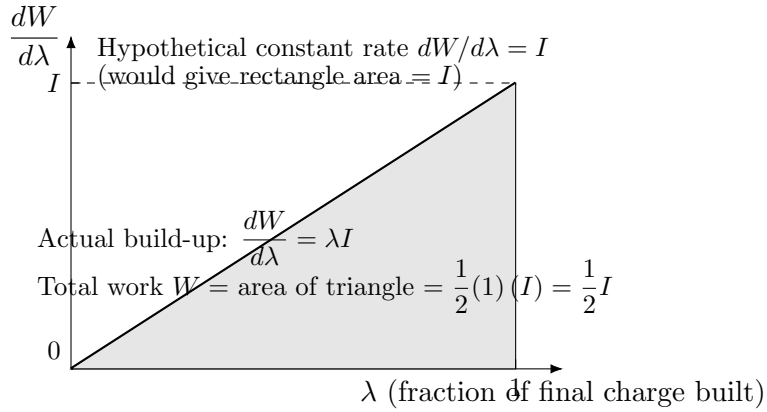


Figure 1: Why the energy has a factor $\frac{1}{2}$: during the gradual build-up, the “work-rate” increases linearly from 0 to I , so the total work equals the triangle area.

One-sentence takeaway

If the charge were added while the full potential ϕ already existed, you would get $W = I$ (rectangle); but because the potential itself grows from 0 to ϕ during construction, the *average* potential is $\phi/2$, giving $W = \frac{1}{2}I$ (triangle).

Example: Energy to assemble four equal point charges on a square

Problem

Four identical point charges, each of magnitude q , are placed at the corners of a square of side length d (in free space). Find the total energy required to *assemble* this configuration from infinity.

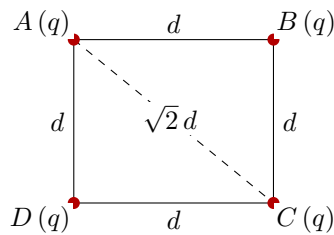


Figure 2: Four equal charges q at the corners of a square of side d .

Method 1: “Potential-at-a-charge” approach

A convenient formula for assembly energy is

$$W = \frac{1}{2} \sum_{i=1}^N q_i \phi_{(\text{due to all other charges})}(\mathbf{r}_i),$$

where the factor $\frac{1}{2}$ avoids double counting pair interactions.

Let us compute the potential at corner A due to the *other three* charges. Distances from A :

$$AB = d, \quad AD = d, \quad AC = \sqrt{2}d.$$

Hence,

$$\phi_A = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{d} + \frac{q}{d} + \frac{q}{\sqrt{2}d} \right) = \frac{q}{4\pi\epsilon_0 d} \left(2 + \frac{1}{\sqrt{2}} \right).$$

By symmetry,

$$\phi_A = \phi_B = \phi_C = \phi_D \equiv \phi.$$

Therefore,

$$W = \frac{1}{2} (4q) \phi = 2q \phi = 2q \cdot \frac{q}{4\pi\epsilon_0 d} \left(2 + \frac{1}{\sqrt{2}} \right) = \frac{q^2}{4\pi\epsilon_0 d} (4 + \sqrt{2}).$$

Answer

$$W = \frac{q^2}{4\pi\epsilon_0 d} (4 + \sqrt{2})$$

Sanity check (pairwise energy sum)

Equivalently, the total energy is the sum over all distinct pairs:

$$W = \sum_{i < j} \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}}.$$

Here there are:

- 4 edge pairs at separation d ,
- 2 diagonal pairs at separation $\sqrt{2}d$.

So,

$$W = \frac{1}{4\pi\epsilon_0} q^2 \left(4 \cdot \frac{1}{d} + 2 \cdot \frac{1}{\sqrt{2}d} \right) = \frac{q^2}{4\pi\epsilon_0 d} (4 + \sqrt{2}),$$

which matches the result above.